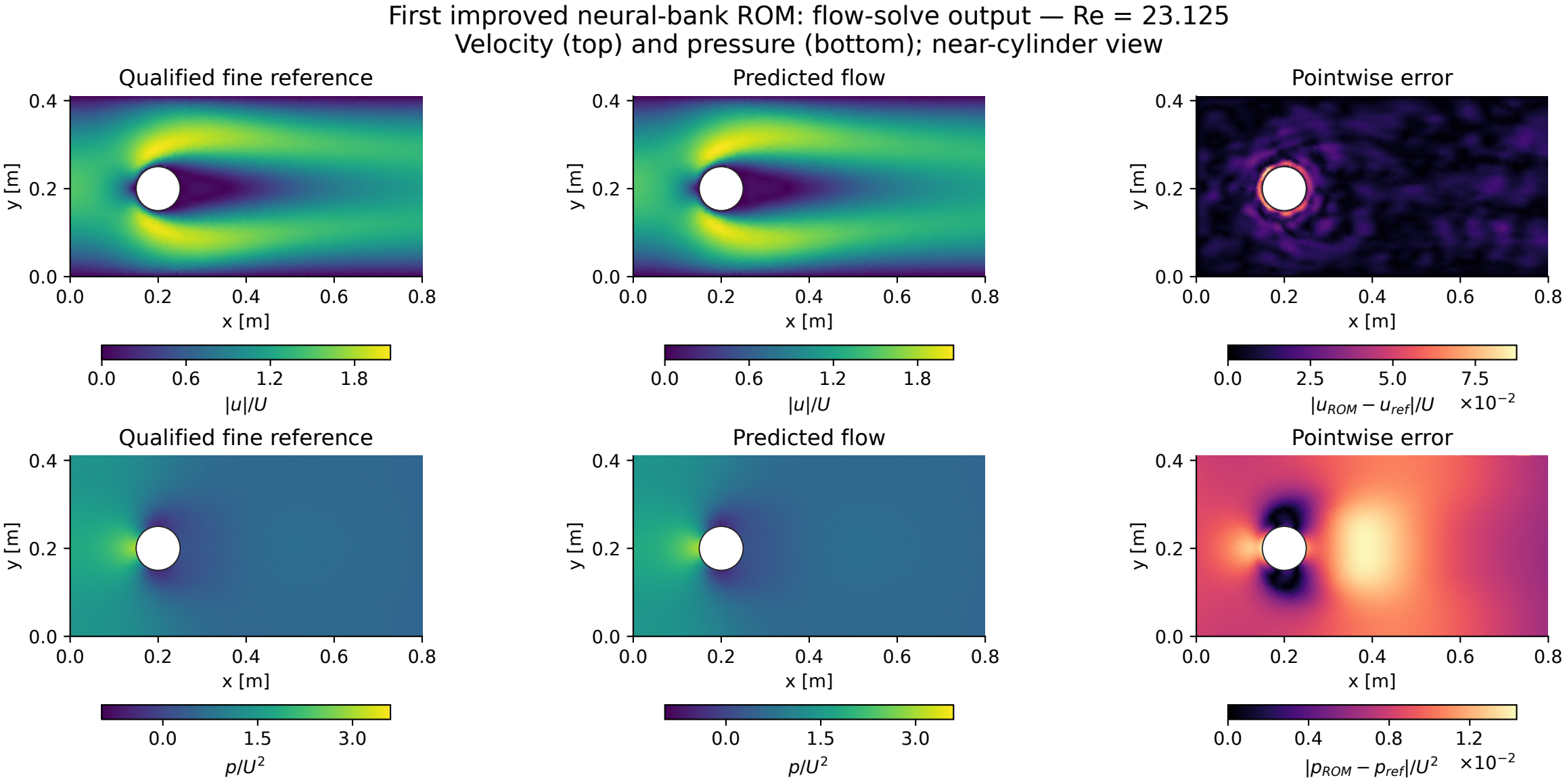

First improved neural-bank ROM: flow-solve output — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Predicted flow
Pointwise error
y [m]
x [m]
|u|/U
|u_ROM − u_ref|/U
×10^−2
p/U^2
|p_ROM − p_ref|/U^2
Four POD-guided neural velocity modes plus FE pressure/stability spaces. Lift, divergence and mass targets still fail.
Display interpolation; error scales differ. Errors against stored reference do not certify equal continuum accuracy. Development cases.